# THE CONGRUENCE $2H_{p-3} \equiv 3 \pmod p$ AND THE INTEGRALITY OF THE TERMS OF OEIS A091817

MATH0

Abstract. For a prime $p \geq 3$ write the harmonic number $H_{p-3} = 1 + \frac{1}{2} + \cdots + \frac{1}{p-3}$ in lowest terms as $N/D$ (with $H_0 = 0/1$). We prove that $p$ divides $2N - 3D$, which confirms Leroy Quet's conjecture, recorded in OEIS A091817 in 2004, that the terms $(2N - 3D)/p$ of that sequence are always integers. The result is not new mathematics: it is the case $k = 3$ of the known congruence $H_{p-k} \equiv H_{k-1} \pmod p$, an immediate consequence of the classical fact $H_{p-1} \equiv 0 \pmod p$, although we found no place where this consequence is recorded for A091817. We classify the note as the first formal proof of a known result. The proof is verified in Lean 4 against a statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against all 18 listed terms and which was reviewed for fidelity and frozen before the proof attempt.

## 1. Introduction

Let $H_m = \sum_{k=1}^{m} 1/k$ denote the $m$-th harmonic number, with $H_0 = 0$. The OEIS sequence A091817 [15], contributed by Mohammed Bouayoun and Ralf Stephan on March 7, 2004, and proposed by Leroy Quet, is

$$a(n) = \frac{2\,\mathrm{num}(H_{p-3}) - 3\,\mathrm{den}(H_{p-3})}{p}, \qquad p \text{ the } (n+1)\text{-th prime}, \ n \geq 1,$$

where num and den denote the numerator and the (positive) denominator in lowest terms. Its first terms, for $p = 3, 5, 7, 11, 13$, are $-1, 0, 2, 62, 554$. The entry records that Quet conjectured that the terms are always integral, and that this was checked for $p < 3 \cdot 10^4$. The math0 referees read the edit history of the entry: the current revision (#8, March 11, 2014) still states the conjecture without a proof or a reference.

**Prior work.** The conjecture follows at once from classical facts. For an odd prime $p$, pairing $1/k$ with $1/(p-k)$ shows that $p$ divides the numerator of $H_{p-1}$; Wolstenholme proved in 1862 that $p^2$ divides it when $p > 3$ [10], [3, Theorem 115]. Since $1/(p-j) \equiv -1/j \pmod p$, this gives the reflection congruence

$$H_{p-k} = H_{p-1} - \sum_{j=1}^{k-1} \frac{1}{p-j} \equiv H_{k-1} \pmod p, \qquad 1 \leq k \leq p-1, \tag{1}$$

understood in the ring of rationals whose denominator is prime to $p$. The congruence in (1) is (2.1) in Lemma 2.1 of Z.-W. Sun's preprint arXiv:0911.4433v3 (2009) [7], whose journal version is [8], and the same derivation appears in the proof of the second congruence in (2.2) of Sun and Zhao [9], the final version of the same arXiv number. The congruence is quoted from [7, 8], in the equivalent form $H_{p-k-1} \equiv H_k \pmod p$ for $1 \leq k \leq p-2$, as Lemma 2.2 of Meštrović [5], and attributed to Sun [8] by Yang and Wang [11]. Taking $k = 3$ gives $H_{p-3} \equiv H_2 = 3/2 \pmod p$, which for $p \geq 5$ is exactly Quet's conjecture; the case $p = 3$ is $a(1) = -1$. The analogous case $k = 2$, namely that $p$ divides $\mathrm{num}(H_{p-2}) - \mathrm{den}(H_{p-2})$, is recorded by Alexander Adamchuk

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(June 9, 2006) in the entry A064169 [14]. We found no place, in the OEIS or elsewhere, where the case $k = 3$ is connected with A091817.

The conjecture is not among the OEIS statements of the formal-conjectures repository [2] (checked at commit `c77fdd4`), nor among the OEIS conjectures of the OEIS Open benchmark [1], and the referees found no other formal proof. What this note adds is therefore a self-contained proof that matches a kernel-checked Lean proof of a frozen formal statement of the conjecture (Section 4); to our knowledge it is the first formal proof of the statement.

## 2. Statement

For $x \in \mathbb{Q}$ we write $x = \mathrm{num}(x)/\mathrm{den}(x)$ in lowest terms with $\mathrm{den}(x) \geq 1$; thus $\mathrm{num}(0) = 0$ and $\mathrm{den}(0) = 1$.

**Definition 1.** For $m \in \mathbb{N}$ let $H_m = \sum_{k=1}^{m} 1/k \in \mathbb{Q}$, and for a prime $p \geq 3$ let

$$t(p) = \frac{2\,\mathrm{num}(H_{p-3}) - 3\,\mathrm{den}(H_{p-3})}{p} \in \mathbb{Q}.$$

For $n \geq 1$ let $a(n) = t(p_n)$, where $p_0 = 2 < p_1 = 3 < p_2 = 5 < \cdots$ are the primes, indexed from 0.

So $a(n)$ is the term for the $(n+1)$-th prime, and $a(1) = t(3)$, in agreement with the offset 1 of the entry and with its PARI program, which loops over the primes $p \geq 3$. The entry's name and the conjecture, an unsigned comment in the entry submitted by M. Bouayoun and R. Stephan on March 7, 2004 [15], read verbatim:

> Name: `(2*numerator(H(p-3)) -3*denominator(H(p-3)))/p with H(m) = 1 + 1/2 + 1/3 + ...+ 1/m.`
>
> Comment: `Leroy Quet conjectures that the terms are always integral. This is true for p < 3*10^4.`

**Theorem 2.** *Let $p \geq 3$ be prime and $H_{p-3} = N/D$ in lowest terms. Then $p \mid 2N - 3D$. Consequently $a(n) \in \mathbb{Z}$ for every $n \geq 1$.*

For $p \geq 5$ the denominator $D$ divides $\mathrm{lcm}(1, \ldots, p-3)$, so $p \nmid D$ and the theorem says $H_{p-3} \equiv 3/2 \pmod{p}$. For $p = 3$ we have $N = 0$, $D = 1$ and $a(1) = -3/3 = -1$.

**Lean formalization.** The statement was formalized by math0 in the file `lean/Math0/Statements/OEIS/A091817.lean`, reviewed for fidelity, and frozen before any proof attempt (its SHA-256 hash begins with `0debd9c5`). Its definitions and the statement are, verbatim except that docstrings, an auxiliary lemma `firstPrimeFrom_spec`, the proof of `primeAt_eq_nth` and one `example` are omitted:

```
namespace Math0.Statements.A091817

def firstPrimeFrom : ℕ → ℕ → ℕ
  | 0, _ => 0
  | fuel + 1, q => if q.Prime then q else firstPrimeFrom fuel (q + 1)

def nextPrime (m : ℕ) : ℕ := firstPrimeFrom (m + 2) (m + 1)

def primeAt : ℕ → ℕ
  | 0 => 2
  | i + 1 => nextPrime (primeAt i)

theorem primeAt_eq_nth (i : ℕ) : primeAt i = Nat.nth Nat.Prime i
```

```
def H (m : ℕ) : ℚ := ((List.range m).map (fun k => (1 : ℚ) / ((k : ℚ) + 1))).sum

def term (p : ℕ) : ℚ := (2 * (H (p - 3)).num - 3 * ((H (p - 3)).den : ℤ) : ℚ) / (p : ℚ)

def a (n : ℕ) : ℚ := term (primeAt n)

theorem a_data : (List.range 18).map (fun i => a (i + 1)) =
    [-1, 0, 2, 62, 554, 74258, 142682, 2830946, 1448977246, 12576415426, 1857881203154,
      64634117421826, 62771945063582, 164497840779865642, 350742295126606034006,
      5774653629556529218142, 5647762835059481932498, 12720214895833193424471634] := by
  decide +kernel

def statement : Prop :=
  ∀ n : ℕ, 1 ≤ n → ∃ z : ℤ, a n = (z : ℚ)

end Math0.Statements.A091817
```

The encoding matches Definition 1. `H m` is the exact rational sum $\sum_{k=0}^{m-1} 1/(k+1) = H_m$, so `H 0` $= 0$, as for the empty sum in PARI. Mathlib's `Rat.num` and `Rat.den` are the numerator and the positive denominator in lowest terms, so they agree with PARI's `numerator` and `denominator`. `term p` is $t(p)$, computed in $\mathbb{Q}$ without rounding; the sequence is $\mathbb{Q}$-valued because its integrality is exactly what the conjecture asserts. `primeAt i` is the $i$-th prime, indexed from 0, computed by a bounded search for the next prime in $(m, 2m+2]$; the lemma `primeAt_eq_nth` proves that it equals Mathlib's `Nat.nth Nat.Prime i`, using Bertrand's postulate. Hence `a n` $= a(n)$. The statement quantifies over $n \geq 1$, that is, over the primes $p \geq 3$, and asserts that $a(n)$ is equal to (the image of) an integer. For $n \geq 1$ the subtraction $p - 3$ in $\mathbb{N}$ is exact and the division is by $p \neq 0$. The restriction $n \geq 1$ is necessary: for $n = 0$, $p = 2$, the truncated subtraction gives $2 - 3 = 0$ and `a 0` $= -3/2$, which the fidelity review checked in the kernel.

The theorem `a_data`, proved by `decide +kernel`, checks that the definitions reproduce all 18 terms listed in the OEIS, $a(1), \ldots, a(18)$ for $p = 3, \ldots, 67$; the omitted `example` checks, also in the kernel, that $a(1), \ldots, a(4)$ belong to $p = 3, 5, 7, 11$. The fidelity reviewer also recomputed the 18 terms independently in Python. The OEIS formula line states $a(n) = (2A001008(p-3) - 3A002805(p-3))/p$ for primes $p \geq 5$ [12, 13]; the data, the PARI program and the formal statement also include $p = 3$, so the formal statement is at least as strong as the conjecture under either reading.

## 3. Proof

We follow the Lean proof, which has four auxiliary results (Lemmas 3–6) and a final deduction. Throughout, $p$ is a prime, $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, and $\overline{x} \in \mathbb{F}_p$ is the residue of an integer $x$. As in Lean, the inverse in the field $\mathbb{F}_p$ is extended by $0^{-1} = 0$; with this convention $x \mapsto x^{-1}$ is an involution of $\mathbb{F}_p$, hence a bijection. For $m \in \mathbb{N}$ put

$$S(m) = \sum_{k=0}^{m-1} \left(\overline{k} + 1\right)^{-1} = \sum_{k=1}^{m} \overline{k}^{\,-1} \in \mathbb{F}_p.$$

This is the private definition `invSum p m`. For $m < p$ every $\overline{k}$ with $1 \leq k \leq m$ is nonzero, and $S(m)$ is the reduction of $H_m$ modulo $p$; the lemmas below make this precise without using localizations.

**Lemma 3** (`H_succ`)**.** *$H_{m+1} = H_m + 1/(m+1)$ for all $m \in \mathbb{N}$.*

*Proof.* This is the definition of $H$ as a sum over $\{0, \ldots, m\}$, split off at the last term. □

**Lemma 4** (`key`)**.** *For every $m < p$ there is an integer $A_m$ with $H_m \cdot m! = A_m$ and $\overline{A_m} = \overline{m!} \cdot S(m)$ in $\mathbb{F}_p$.*

*Proof.* Induction on $m$. For $m = 0$ take $A_0 = 0$: $H_0 \cdot 0! = 0$ and $S(0) = 0$. Now let $m + 1 < p$ and let $A_m$ be given by the induction hypothesis (applicable since $m < p$). Put $A_{m+1} = A_m(m + 1) + m!$. By Lemma 3,

$$H_{m+1}(m+1)! = \Big(H_m + \frac{1}{m+1}\Big)(m+1)\,m! = H_m m!\,(m+1) + m! = A_{m+1}.$$

Modulo $p$, since $1 \le m+1 < p$, the residue $\overline{m+1} = \overline{m}+1$ is nonzero, so $(\overline{m}+1)(\overline{m}+1)^{-1} = 1$. Using $S(m+1) = S(m) + (\overline{m}+1)^{-1}$ and the induction hypothesis,

$$\overline{(m+1)!}\,S(m+1) = (\overline{m}+1)\,\overline{m!}\,S(m) + \overline{m!}\,(\overline{m}+1)(\overline{m}+1)^{-1} = \overline{A_m}(\overline{m}+1) + \overline{m!} = \overline{A_{m+1}}. \quad \square$$

**Lemma 5** (`invSum_self`)**.** *If $p \neq 2$, then $S(p) = \sum_{k=1}^{p} \overline{k}^{-1} = 0$.*

*Proof.* First, $\sum_{x\in\mathbb{F}_p} x^{-1} = \sum_{x\in\mathbb{F}_p} x$, because $x \mapsto x^{-1}$ is a bijection of $\mathbb{F}_p$. Second, $\sum_{x\in\mathbb{F}_p} x = 0$: this is the case $i = 1$ of the fact that $\sum_{x\in\mathbb{F}_q} x^i = 0$ for $i < q-1$ in a finite field with $q$ elements (Mathlib's `FiniteField.sum_pow_lt_card_sub_one`), which applies because $1 < p-1$ for $p \ge 3$. (Directly: $\sum_{x=0}^{p-1} x = p \cdot \frac{p-1}{2}$ with $\frac{p-1}{2} \in \mathbb{Z}$.) Third, $k \mapsto \overline{k}$ is a bijection from $\{0, 1, \dots, p-1\}$ onto $\mathbb{F}_p$, with inverse $x \mapsto$ the least nonnegative representative of $x$; hence $\sum_{k=0}^{p-1} \overline{k}^{-1} = \sum_{x\in\mathbb{F}_p} x^{-1} = 0$. Finally, the sum $\sum_{k=0}^{p} \overline{k}^{-1}$ can be split off at either end:

$$\overline{0}^{-1} + \sum_{k=0}^{p-1} (\overline{k}+1)^{-1} = \sum_{k=0}^{p} \overline{k}^{-1} = \sum_{k=0}^{p-1} \overline{k}^{-1} + \overline{p}^{-1}.$$

Since $\overline{0}^{-1} = 0^{-1} = 0$ and $\overline{p} = 0$, so that $\overline{p}^{-1} = 0$, this gives $S(p) = \sum_{k=0}^{p-1} \overline{k}^{-1} = 0$. $\square$

Since $\overline{p}^{-1} = 0$, Lemma 5 is the statement $\sum_{k=1}^{p-1} \overline{k}^{-1} = 0$, i.e. $H_{p-1} \equiv 0 \pmod p$.

**Lemma 6** (`invSum_val`)**.** *If $p \neq 2$, then $2\,S(p-3) = 3$ in $\mathbb{F}_p$.*

*Proof.* Since $p \ge 3$ we may write $p = q + 3$ with $q = p - 3 \in \mathbb{N}$. In $\mathbb{F}_p$ we have $\overline{q} = -3$, hence $\overline{q}+1 = -2$, $\overline{q}+2 = -1$ and $\overline{q}+3 = 0$. Splitting off the last three terms of $S(q+3)$ and using Lemma 5,

$$0 = S(q+3) = S(q) + (-2)^{-1} + (-1)^{-1} + 0^{-1} = S(q) - 2^{-1} - 1.$$

As $p \nmid 2$, $\overline{2} \neq 0$ and $2 \cdot 2^{-1} = 1$. Multiplying by 2 gives $0 = 2S(q) - 1 - 2$, that is, $2S(p-3) = 3$. $\square$

For $p = 3$ this reads $2 \cdot S(0) = 0 = 3$ in $\mathbb{F}_3$, which is consistent.

*Proof of Theorem 2.* We prove the formal statement: for every $n \ge 1$ there is $z \in \mathbb{Z}$ with $a(n) = z$. Let $n \ge 1$ and $p = p_n$. By `primeAt_eq_nth`, $p =$ `Nat.nth Nat.Prime n`, the $n$-th prime counted from 0, so $p$ is prime; since the enumeration of the primes is strictly increasing and $p_0 = 2$, we get $p > 2$, so $p \ge 3$.

Let $m = p-3 < p$ and let $A = A_m$ be the integer of Lemma 4, so $H_m\,m! = A$ and $\overline{A} = \overline{m!}\,S(m)$. Write $H_m = N/D$ in lowest terms. Since $H_m D = N$ (Mathlib's `Rat.mul_den_eq_num`), we have $N \cdot m! = H_m D \cdot m! = A \cdot D$, an identity in $\mathbb{Q}$ between integers and hence in $\mathbb{Z}$. Reducing modulo $p$ gives

$$\overline{N} \cdot \overline{m!} = \overline{m!}\,S(m) \cdot \overline{D}.$$

As $m < p$, the prime $p$ does not divide $m!$, so $\overline{m!} \neq 0$ and we may cancel it: $\overline{N} = S(m)\,\overline{D}$. By Lemma 6,

$$\overline{2N - 3D} = 2S(m)\,\overline{D} - 3\,\overline{D} = (2S(m) - 3)\,\overline{D} = 0.$$

Hence $p \mid 2N - 3D$; write $2N - 3D = pz$ with $z \in \mathbb{Z}$. Since $p \neq 0$ in $\mathbb{Q}$, we get $a(n) = t(p) = (2N - 3D)/p = z$. □

**Remark 7.** The argument does not use that $p \nmid D$: the factorial $m!$, which is prime to $p$, plays the role of a common denominator. For $p \geq 5$ the theorem is equivalent to $H_{p-3} \equiv 3/2 \pmod p$, and Lemma 5 together with $(p-1)^{-1} \equiv -1$ and $(p-2)^{-1} \equiv -1/2$ is the case $k = 3$ of the derivation of (1).

## 4. Machine verification

**Files and declarations.** The proof is in the file `lean/Math0/Frontier/OEIS/A091817.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A091817.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. Its declarations are the following (signatures verbatim, proofs omitted):

```
namespace Math0.Frontier.A091817
open Math0.Statements.A091817

private def invSum (p m : ℕ) : ZMod p := ∑ k ∈ Finset.range m, ((k : ZMod p) + 1)⁻¹

private theorem H_succ (m : ℕ) : H (m + 1) = H m + 1 / ((m : ℚ) + 1)

private theorem key (p : ℕ) [Fact p.Prime] : ∀ m, m < p → ∃ a : ℤ,
    H m * (m.factorial : ℚ) = a ∧ (a : ZMod p) = (m.factorial : ZMod p) * invSum p m

private theorem invSum_self (p : ℕ) [hp : Fact p.Prime] (h2 : p ≠ 2) : invSum p p = 0

private theorem invSum_val (p : ℕ) [hp : Fact p.Prime] (h2 : p ≠ 2) :
    2 * invSum p (p - 3) = 3

theorem result : statement

end Math0.Frontier.A091817

#math0_settles Math0.Frontier.A091817.result Math0.Statements.A091817.statement
```

They correspond to Lemmas 3–6 and Theorem 2. The main declaration is `Math0.Frontier.A091817.result`, of type `Math0.Statements.A091817.statement`. The proofs use, besides the tactics `field_simp`, `push_cast` and `linear_combination`, the Mathlib lemmas `Equiv.sum_comp` with `Equiv.inv` and `FiniteField.sum_pow_lt_card_sub_one` (Lemma 5), `ZMod.natCast_eq_zero_iff` and `Nat.Prime.dvd_factorial` (the residues of $m+1$, 2 and $m!$ are nonzero), `Rat.mul_den_eq_num` (the identity $H_m D = N$), `ZMod.intCast_zmod_eq_zero_iff_dvd` (passing from $\mathbb{F}_p$ back to divisibility in $\mathbb{Z}$), and `Nat.nth_strictMono` with `Nat.nth_prime_zero_eq_two` (the bound $p \geq 3$). The proof involves no finite computation; the only kernel evaluations in this result are the data checks in the statement file (`a_data` and the omitted `example`), which are not used by the proof. Neither file uses `native_decide`, `sorry` or a `maxHeartbeats` option.

**Audit.** The file ends with `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself. It fails unless (1) `statement` is a definition of type `Prop` without universe parameters; (2) `statement` and `result` depend on no axioms other than `propext`, `Classical.choice` and `Quot.sound` (this rejects `sorryAx`, `Lean.ofReduceBool`, `Lean.trustCompiler` and custom axioms); and (3) the type of `result` is syntactically the statement or its negation, after removing metadata annotations, with no unfolding. On this file it reports

```
math0 audit: 'Math0.Statements.A091817.statement' depends on axioms [Classical.choice, Quot.
    sound, propext]
math0 audit: 'Math0.Frontier.A091817.result' depends on axioms [Classical.choice, Quot.sound,
    propext]
math0 statement check: 'Math0.Frontier.A091817.result' proves 'Math0.Statements.A091817.
    statement'
```

**Toolchain and reproduction.** We use Lean 4 [6] `v4.33.1` and Mathlib [4] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [2] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root: it fetches the Mathlib build cache, builds all math0 results with `lake build` (re-running every audit command), checks that the pinned `formal_conjectures` and `mathlib` checkouts are unmodified, and reports the number of passed statement checks. The single file can be checked with `lake env lean Math0/Frontier/OEIS/A091817.lean` in the directory `lean`; this takes well under a minute on our machine, most of it spent loading Mathlib.

## References


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[12] The OEIS Foundation Inc., `https://oeis.org/A001008` (numerators of harmonic numbers).

[13] The OEIS Foundation Inc., `https://oeis.org/A002805` (denominators of harmonic numbers).

[14] The OEIS Foundation Inc., `https://oeis.org/A064169`, entry by L. Quet (2001), comment by A. Adamchuk (2006).

[15] The OEIS Foundation Inc., `https://oeis.org/A091817`, entry by M. Bouayoun and R. Stephan (2004), proposed by L. Quet.


Produced by Claude (an AI model developed by Anthropic) in the math0 project